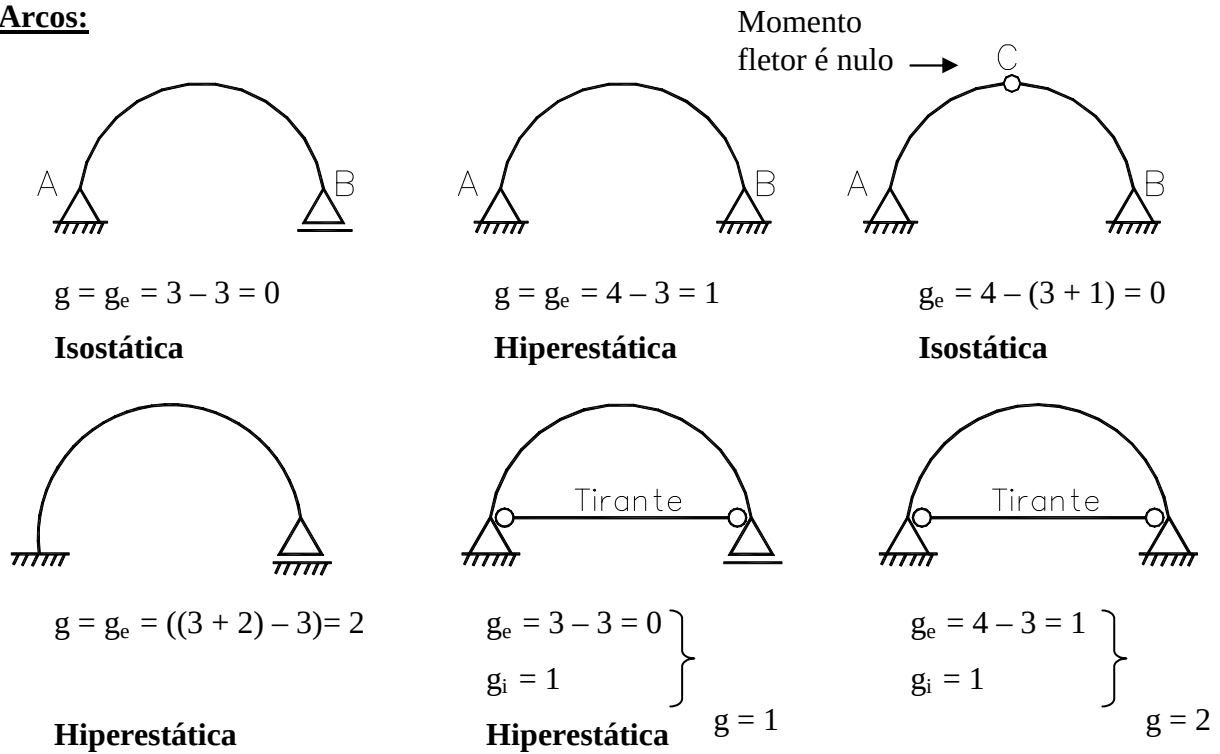
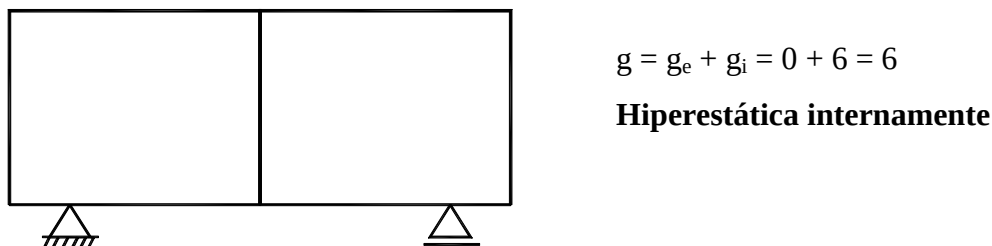
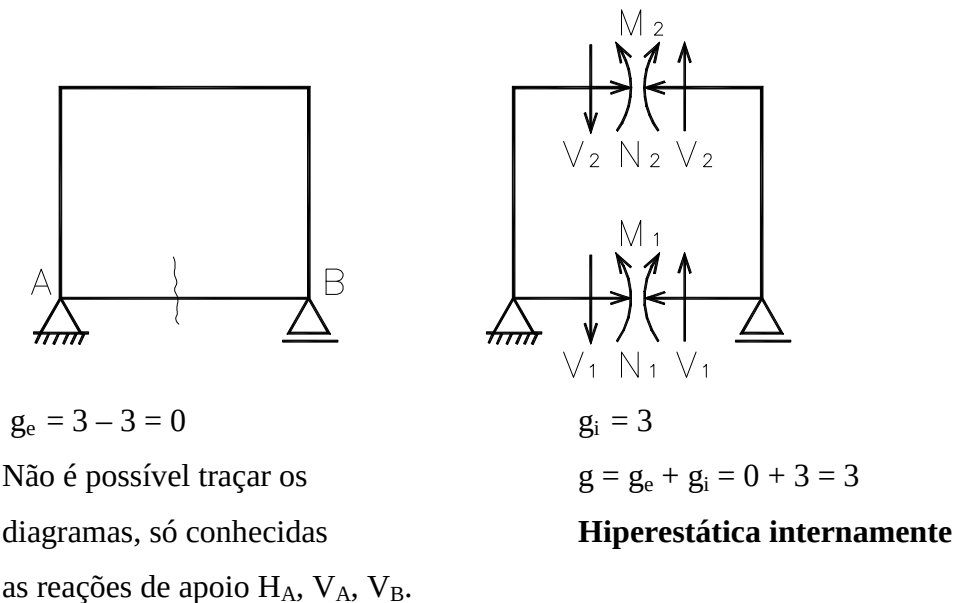


Arcos:



Quadros:

Conhecidos N_1 , V_1 e M_1 obtêm-se os esforços N_2 , V_2 e M_2 ou em qualquer seção.

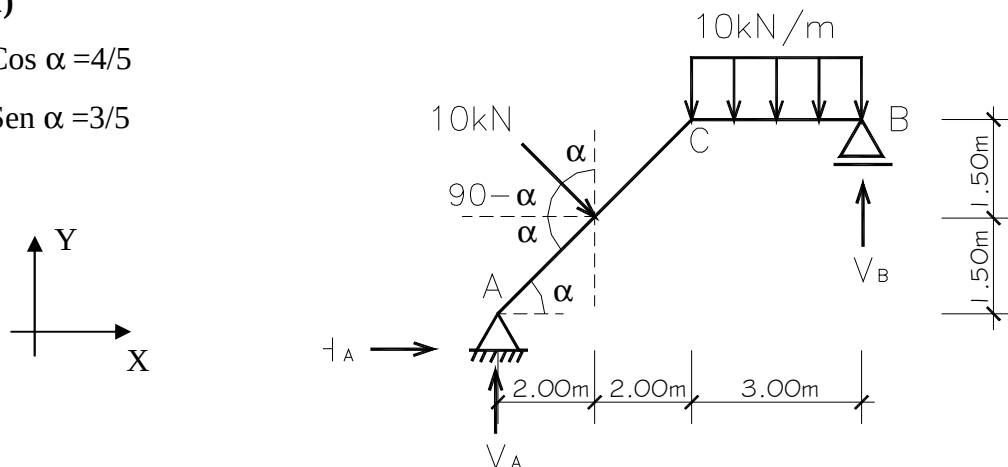


1.5 – Reações de apoio em estruturas planas:

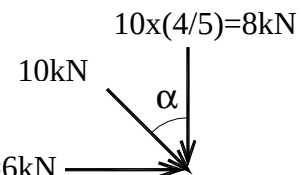
1)

$$\cos \alpha = 4/5$$

$$\sin \alpha = 3/5$$



Decompor a força de 10kN nas direções x e y:



$$\text{i) } \sum F_X = 0 \quad H_A + 6 \text{ kN} = 0 \quad \therefore H_A = -6 \text{ kN}$$

$$\text{ii) } \sum F_Y = 0 \quad V_A + V_B = (10 \times 3) + 8 = 38 \text{ kN}$$

$$\text{iii) } \sum M_A = 0 \quad 7 \times V_B - (30 \times 5,5) - (8 \times 2) - (6 \times 1,5) = 0$$

$$\therefore 7V_B = 190 \quad \therefore V_B = 27,14 \text{ N}$$

$$\text{Logo, } V_A = 38 \text{ kN} - 27,14 \text{ kN} = 10,86 \text{ kN}$$

Outra maneira seria:

$$\sum M_A = 0$$

$$7V_B - (30 \times 5,5) - (10 \times 2,5) = 0$$

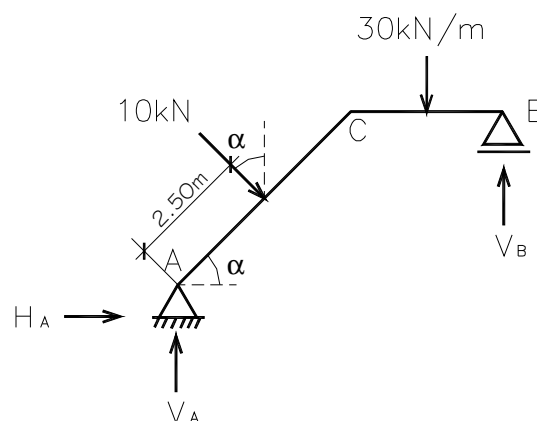
$$\therefore 7V_B = 165 + 25 = 190$$

$$\therefore V_B = 27,14 \text{ kN}$$

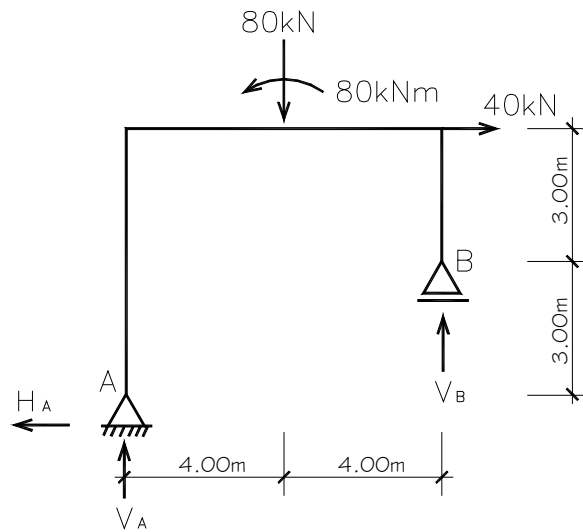
$$\text{Verificação: } \sum M_B = 0$$

$$(10,86 \times 7) + (6 \times 3) - (30 \times 1,5) - (8 \times 5) = 0$$

$$76 + 18 - 45 - 40 - 9 = 0$$



2)



$$\begin{aligned} \text{i) } \sum F_X = 0 & \quad -H_A + 40 = 0 \quad \therefore H_A = 40\text{kN} \\ \text{ii) } \sum F_Y = 0 & \quad V_A + V_B = 60\text{kN} \\ \text{iii) } \sum M_A = 0 & \quad 8V_B + 80 - (40 \times 6) - (60 \times 4) = 0 \\ & \quad \therefore 8V_B = 400 \quad \therefore V_B = 50\text{kN} \\ & \quad \therefore V_A = 60 - 50 = 10\text{kN} \end{aligned}$$

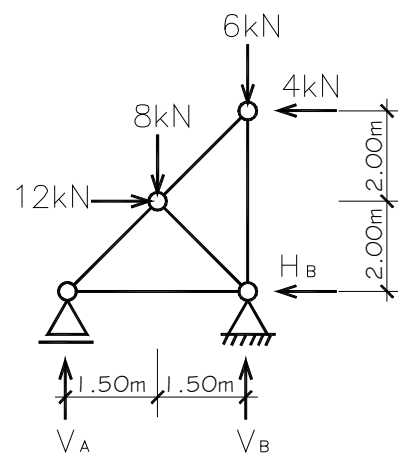
Verificação: $\sum M_B = 0$

$$(10 \times 8) + (40 \times 3) - 80 - (60 \times 4) + (40 \times 3) = 0$$

$$120 + 120 - 240 = 0$$

3)

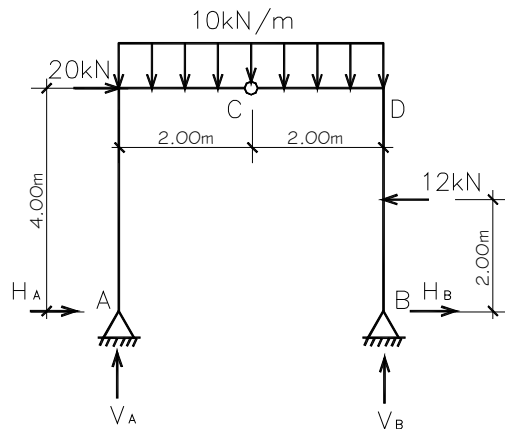
$$\begin{aligned} \text{i) } \sum F_X = 0 & \quad H_B + 4 - 12 = 0 \quad \therefore H_B = 8\text{kN} \\ \text{ii) } \sum F_Y = 0 & \quad V_A + V_B = 6 + 8 = 14\text{kN} \\ \text{iii) } \sum M_B = 0 & \quad (4 \times 4) + (8 \times 1,5) - (12 \times 2) - 3V_A = 0 \\ & \quad \therefore 3V_A = 16 + 12 - 24 = 4 \\ & \quad \therefore V_A = (4/3) = 1,33\text{kN} \\ & \quad \therefore V_B = 12,67\text{kN} \end{aligned}$$



Verificação: $\sum M_A = 0$

$$\begin{aligned} r=3; b=5; n=4. & \quad \left\{ \begin{array}{l} r + b = 2n \\ 5 + 3 = 2 \times 4 \end{array} \right. \end{aligned}$$

4) Pórtico Tri-articulado



- 4 Incógnitas (Reação)
- 3 Equações Estáticas
- 1 Equação interna
- $M_C^D = M_C^E = 0$

Isostática

i) $\sum F_X = 0$ $H_A + H_B + 20 - 12 = 0 \quad \therefore H_A + H_B = -8\text{kN}$

ii) $\sum F_Y = 0$ $V_A + V_B = 10 \times 4 = 40\text{kN}$

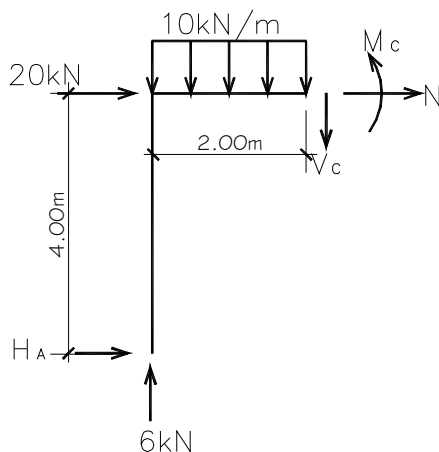
iii) $\sum M_A = 0$ $4V_B - (40 \times 2) + (12 \times 2) - (20 \times 4) = 0$



$\therefore 4V_B = 80 - 24 + 80 \quad \therefore V_B = 34\text{kN}$

$\therefore V_A = 40 - 34 = 6\text{kN}$

iv) Momento Fletor em C é nulo (Esq. Ou Dir.)



$M_C - (6 \times 2) + (20 \times 1) + (H_A \times 4) = 0$

Ou $M_C = (6 \times 2) - (20 \times 1) - (4H_A)$

Mas $M_C = 0 \rightarrow 4H_A = 12 - 20 = -8$

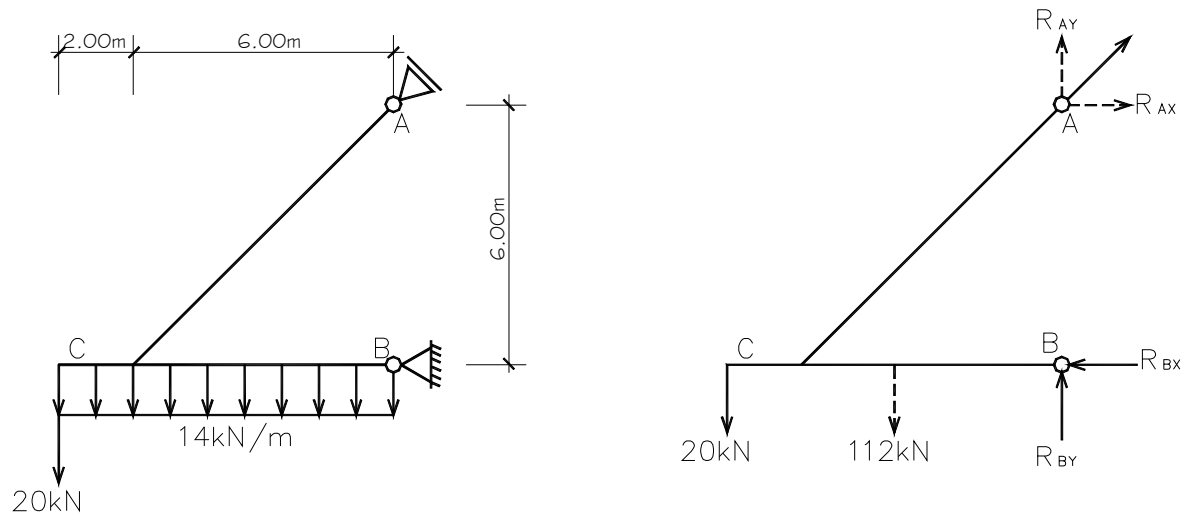
$\therefore H_A = -2\text{kN}$

$\therefore H_B = -8 + 2 = -6\text{kN}$

Verif. $\sum M_D = 0$ $(6 + 2) \times 4 + (12 \times 2) + (6 \times 4) - (40 \times 2) = 0$

$48 + 24 + 24 - 80 = 0$

Determinar a reação de apoio



$$\sum F_X = 0 (\rightarrow^+)$$

$$R_{AX} - R_{BX} = 0$$

$$\therefore R_{AX} = R_{BX} \quad (I)$$

$$\sum F_Y = 0 (\uparrow^+)$$

$$R_{AY} - R_{BY} - 20 - 112 = 0$$

$$\therefore R_{AY} + R_{BY} = 132$$

$$\sum M_A = 0 \quad \curvearrowright^+$$

$$(20 \times 8) + (112 \times 4) - (6 \times R_{BX}) = 0$$

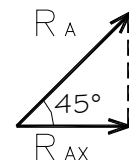
$$R_{BX} = \frac{160 + 448}{6} \quad \therefore R_{BX} = 101,33 \text{ kN}$$

$$R_{AX} = R_{BX} \quad (I) \quad \therefore R_{AX} = 101,33 \text{ kN}$$

$$R_{AX} = R_{BY} \quad (45^\circ) \quad \therefore R_{AY} = 101,33 \text{ kN}$$

$$R_{BY} = 132 - R_{AY} \quad \therefore R_{BY} = 30,67 \text{ kN}$$

$$R_A = R_{AX} / \cos 45^\circ \quad \therefore R_A = \frac{(R_{AX}) \times 2}{\sqrt{2}} = 143,30 \text{ kN}$$



Conferindo

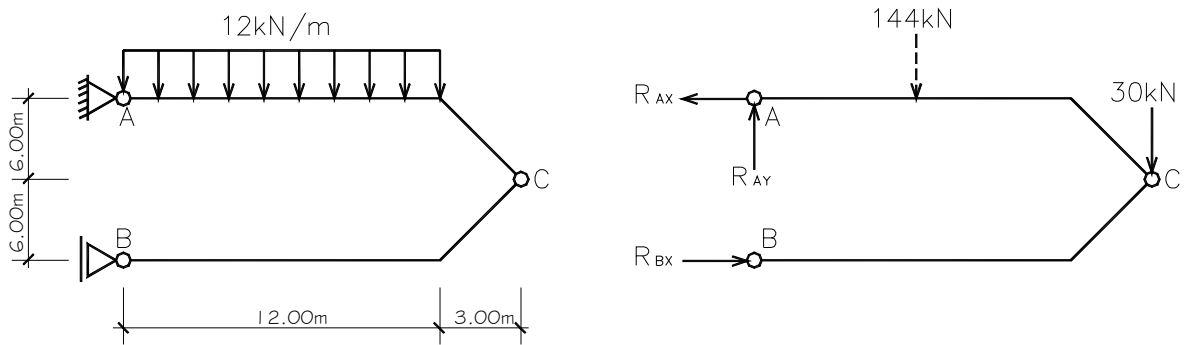
$$\sum M_C = 0 \quad \curvearrowright^+$$

$$(20 \times 2) - (112 \times 2) + (6 \times R_{BY}) - (6 \times R_{AX}) + (6 \times R_{AY}) = 0$$

$$10 - 224 + (30,67 \times 6) - (101,33 \times 6) + (101,33 \times 6) = 0$$

$$-184 + 184 - 608 + 608 = 0$$

$$184 - 184 = 0$$



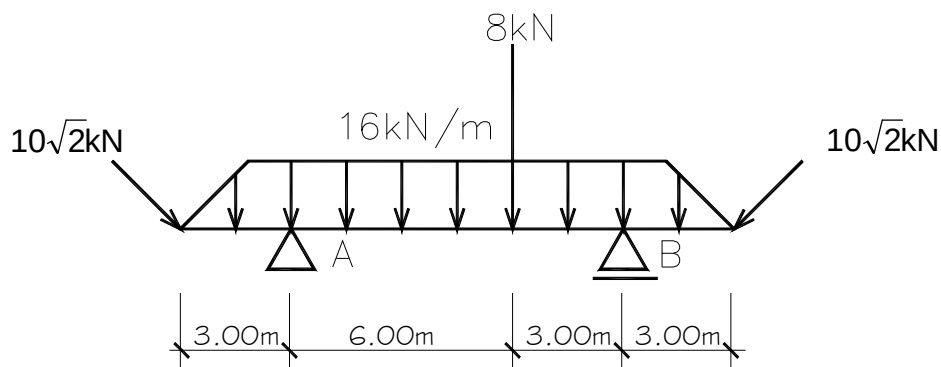
$$\begin{aligned}
 \text{i) } \sum F_X &= 0 & R_{AX} &= R_{BX} \\
 \text{ii) } \sum F_Y &= 0 & R_{AY} - 12(12) - 30 &= 0 & R_{AY} &= 174 \text{ kN} \\
 \text{iii) } \sum M_A &= 0 & 12xR_{BX} - 30x20 - 144x6 &= 0 \\
 & & R_{BX} &= \frac{600 + 864}{12} & R_{BX} &= 122 \text{ kN} & R_{AX} &= 122 \text{ kN}
 \end{aligned}$$

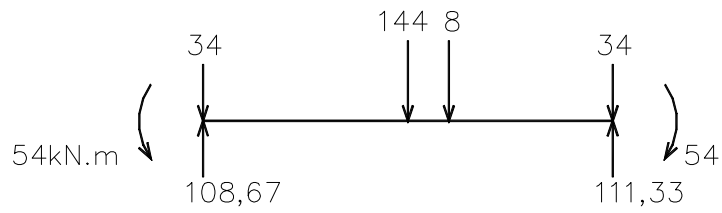
Conferindo

$$\begin{aligned}
 \sum M_B &= 0 & 12xR_{AX} - 144x6 - 30x20 &= 0 \\
 & & 1464 - 864 - 600 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \sum M_C &= 0 & 6xR_{BX} - 144x14 + 6xR_{AX} - 20xR_{AY} &= 0 \\
 & & 122x6 + 2016 + 122x6 - 174x20 &= 0 \\
 & & 732 + 2016 + 732 - 3480 &= 0
 \end{aligned}$$

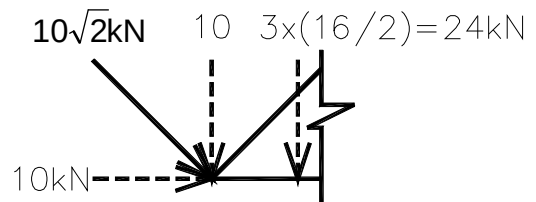
Achar as reações de apoio para a viga abaixo :



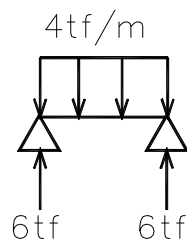
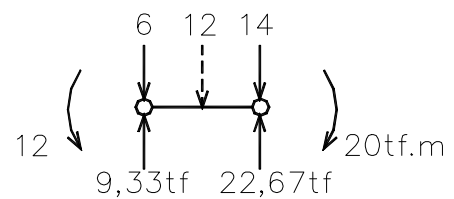
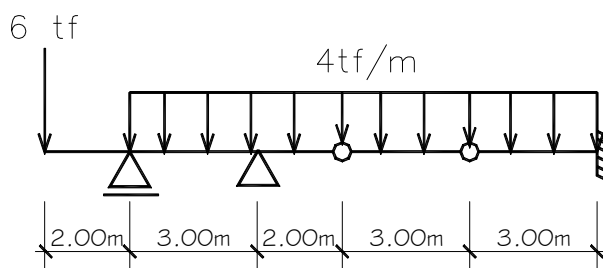


$72 \uparrow$	$\uparrow (144/2) = 72$
$34 \uparrow$	$\uparrow 10 + 24 = 34$
$(8 \times 3)/9 = 2,67 \uparrow$	$\uparrow (8 \times 6)/9 = 5,33$
$108,67 \uparrow$	$\uparrow 111,33$

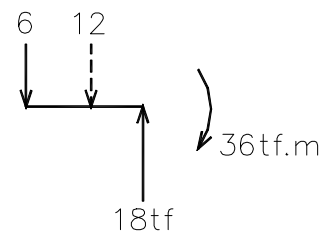
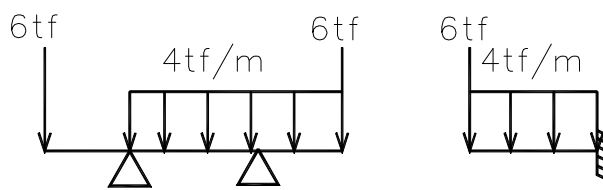
Balanço



Determinar as reações de apoio para a viga:



$6 \uparrow$	$\uparrow (12/2) = 6$
$6 \uparrow$	$\uparrow 6 + 8 = 14$
$2,67 \uparrow$	$\uparrow (20-12)/3 = 2,67$



1.6 – Reações de apoio no espaço:

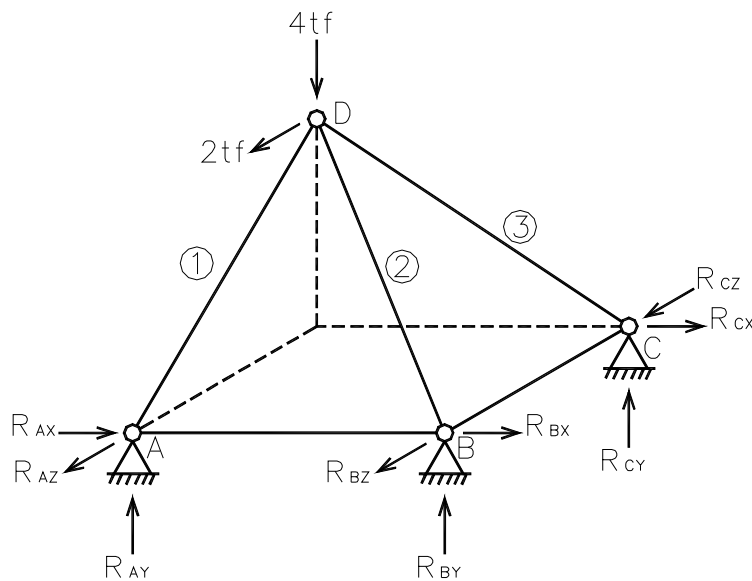
6 Equações de Equilíbrio:

$$\sum F_X = 0; \quad \sum F_Y = 0; \quad \sum F_Z = 0; \quad \sum M_X = 0; \quad \sum M_Y = 0; \quad \sum M_Z = 0$$

1) Treliça

Isostática

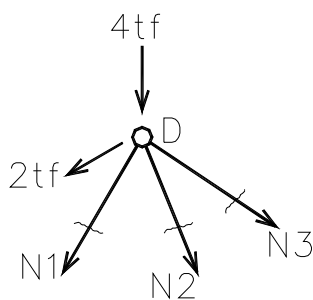
$$\begin{cases} r + b = 2n \\ \text{Restringida} \end{cases}$$



$$\begin{aligned} r &= 3 \times 3 = 9 \\ b &= 3 \\ h &= 4 \end{aligned}$$

$$\begin{aligned} r + b &= 3 \times h \\ 9 + 3 &= 3 \times 4 \\ 12 &= 12 \end{aligned}$$

Inicia-se pelo equilíbrio do nó D:



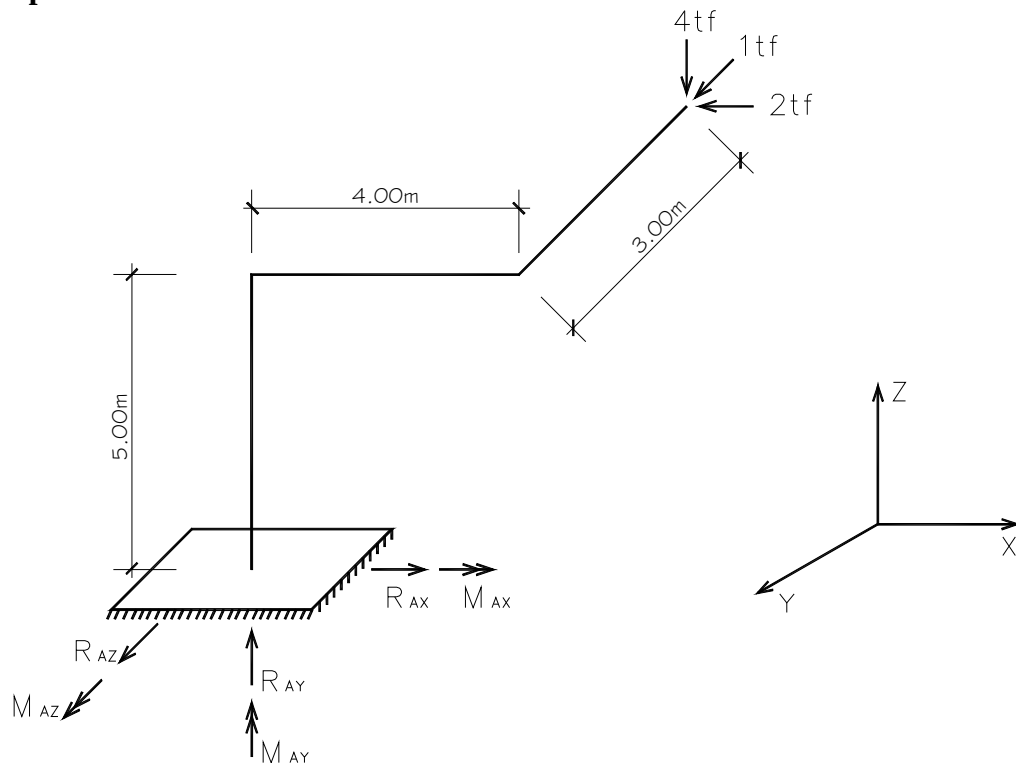
3 incógnitas

N1, N2, N3

3 equações: $\sum F_X = 0$, $\sum F_Y = 0$, $\sum F_Z = 0$

Em seguida passa-se aos nós com apoios: Conhecidos agora os esforços N1, N2 e N3, para cada nó A, B ou C existem 3 incógnitas (Reações) e 3 equações de equilíbrio.

2) Pórtico Espacial



Isostática { 6 reações
 6 equações de equilíbrio
 Restringida

i) $\sum F_X = 0$	$R_{AX} - 2tf = 0$	$\therefore R_{AX} = 2tf$
ii) $\sum F_Y = 0$	$R_{AY} - 4tf = 0$	$\therefore R_{AY} = 4tf$
iii) $\sum F_Z = 0$	$R_{AZ} - 1tf = 0$	$\therefore R_{AZ} = 1tf$
iv) $\sum M_X = 0$	$M_{AX} - (4 \times 3) - (1 \times 5) = 0$	$\therefore M_{AX} = 17tfm$
v) $\sum M_Y = 0$	$M_{AY} + (2 \times 3) + (1 \times 4) = 0$	$\therefore M_{AY} = -10tfm$
vi) $\sum M_Z = 0$	$M_{AZ} + (2 \times 5) - (4 \times 4) = 0$	$\therefore M_{AZ} = 6tfm$